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In Blessed Memory of Aleksei Nikitich Khramov

Paleomagnetism of Carboniferous Formations of the Russian Platform: Reinvestigation of Old Collections

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Abstract—The collections of Carboniferous rocks from sections of the Russian Platform (Gzhelian, Moscovian, Bashkirian, and Visean stages) are studied. The new mean paleomagnetic poles are obtained from the Gzhelian, Moscovian, and Visean layers of the Carboniferous of the Russian Platform. In the redbed Gzhelian and Moscovian rocks, the natural remanent magnetization (NRM) components with the inclination shallowing are revealed, which is due to the presence of the large hematite particles or particle aggregates associated with the interaction between the magnetic and clay particles. Based on the obtained determinations and the results contained in the World paleomagnetic database, the trajectory of the apparent polar wander path (APWP) for the East European Platform is constructed in the interval from the Devonian to Early Permian. The Carboniferous kinematics of the East European Platform is estimated.

Keywords: paleomagnetic poles, Carboniferous, anisotropy of magnetic susceptibility, Russian Platform

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INTRODUCTION

The Carboniferous Period has a duration of 60 Ma (*Stratigraficheskii*..., 2006; *Dopolneniya*..., 2000; *The Geological*..., 2012). The paleomagnetic studies of this period in the evolution of the Earth revealed a number of interesting features. This time was marked with the progressive northward advance of the Russian Platform from the near-equatorial region to the midlatitudes. According to different estimates, the rate of the motion in the Carboniferous reached 5–10 cm per annum. The behavior of the geomagnetic field has changed from the state characterized by frequent reversals to the predominance of the reversed polarity—the beginning of the Kiama hyperchron (C_2–P_2). Among the other peculiarities is the presence of mass extinctions of plants and animals at the Devonian/Carboniferous and Lower/Upper Carboniferous boundaries (the Famennian/Tournaisian and Serpukhovian/Bashkirian mass extinction events (Alekseev, 2000)). Understanding these features of the Carboniferous requires detailed information about the behavior of the geomagnetic field during this geological period. Obtaining this information will permit a more comprehensive interpretation of the processes that took place in the Earth's interior and on the surface.

The paleomagnetic studies of the Carboniferous formations of the Russian Platform were started at the end of the 1950s and the beginning of the 1960s under the direction of A.N. Khramov at the paleomagnetic laboratory of the All-Russia Petroleum Research Exploration Institute (VNIGRI) (Kalashnikov, 1961; Khramov and Komissarova, 1963; Khramov, 1963a; 1963b). In the 1950s–1970s, more than 50 results were obtained from the Carboniferous rocks of the Russian Platform (*Paleomagnitnye*..., 1971; 1973; 1975; 1979; 1982; *Dannye*..., 1984). The quality of these determinations does not always meet the criteria of confidence estimates, which requires both the reinvestigation of the existing collections and studying the new collections of the Carboniferous rocks. Another difficulty is associated with the fact that most of the results are obtained from red rocks, in which case the record of the geomagnetic field variations can be delayed up to the complete remagnetization of the rocks during the subsequent epochs. The analysis of the results obtained by the paleomagnetic studies of the Carboniferous rocks of the East European Platform demonstrates the incompleteness of the existing time series of paleomagnetic determinations. The Early Carboniferous, which has a duration of ~40 Ma, is covered by half as many determinations meeting the present-day reliability criteria as the epochs of the Late Carboniferous (18 Ma). In this paper, we present the results for the collections that were gathered by Khramov and other

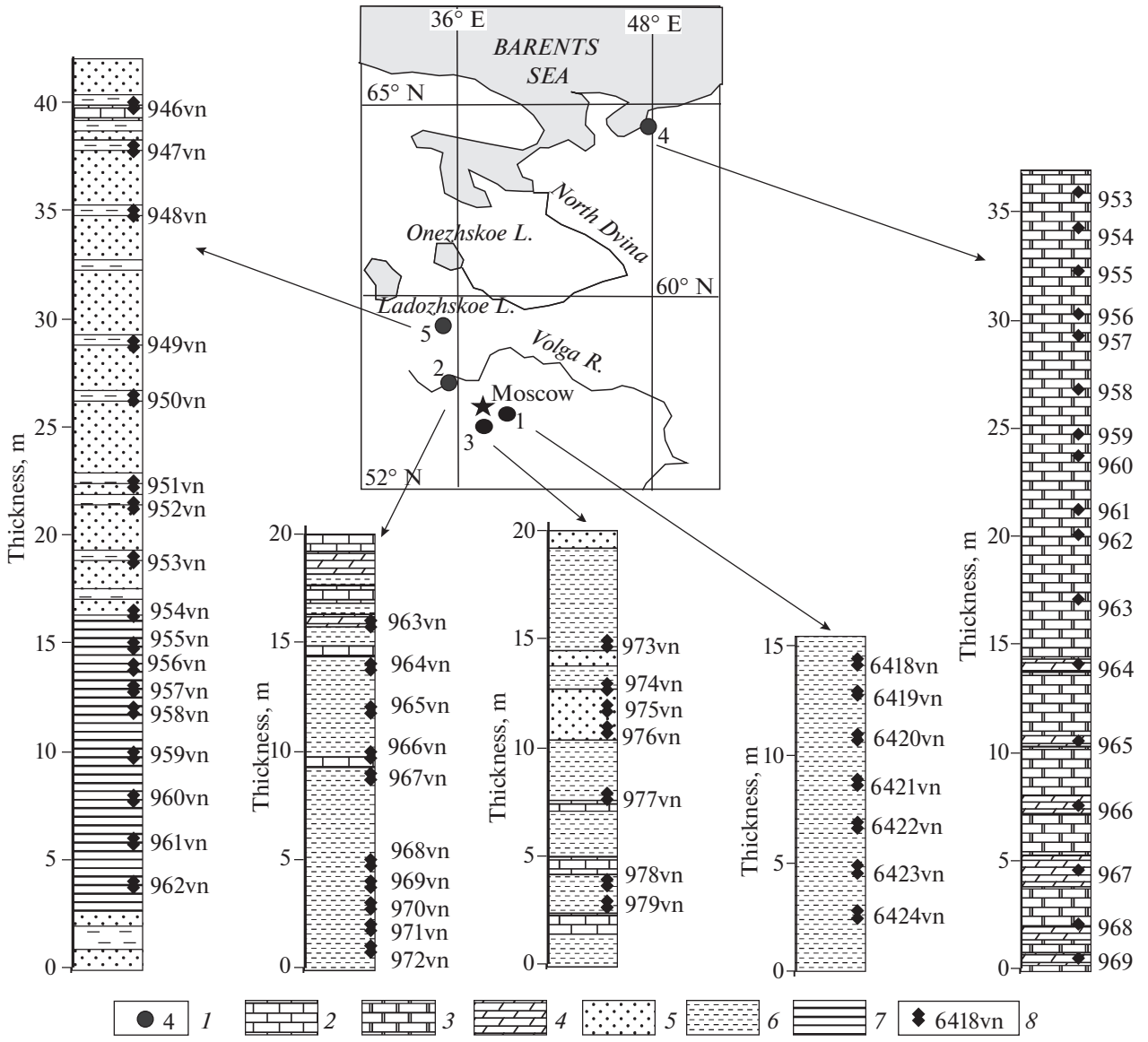


Fig. 1. The schematic location of the studied sections and sampling sites: (*1*) sections: 1, Gzhel; 2, River Volga (Rzhev), 3, River Nara (Pushchino), 4, River Volonga, 5, Boksitogorsk; rock types: (*2*) limestones; (*3*) dolomites; (*4*) dolomitized marls; (*5*) sands; (*6*) clays; (*7*) bauxites; (*8*) sampling sites.

scientists of the paleomagnetic laboratory of VNIGRI in the 1950s–1970s from the Carboniferous rocks of the Russian Platform and North Timan.

OBJECTS OF STUDY AND METHODS OF ANALYSIS

Moscow Region, Gzhel site. The collection of rocks from the Shchelkovo series of the Gzhel stage of the Carboniferous (Khramov's collection, 1959) was gathered in the region of the Gzhel station in a quarry. In the present paper, greenish brown and brick-red clays representing a 15-m interval of the sequence with 14 stratigraphic levels are investigated (*Geologiya*..., 1971). The bedding is close to horizontal and the sampling site has the coordinates 55.61° N, 38.45° E, Fig. 1.

Pushchino locality, right bank of the Nara river (a tributary of the Oka river, Khramov's collection, 1961). Brownish red clays of the Vereyan horizon, Moscovian stage of the Carboniferous. The total thickness of the sampled sequence is 15 m, 14 stratigraphic levels (*Geologiya*..., 1971). The bedding is close to horizontal, sampling site 54.94° N, 37.35° E, Fig. 1.

Tver Oblast, town of Rzhev (Khramov's collection, 1961). The Rzhev layers of the Kashira horizon (Moscovian stage of the Barboniferous) are sampled on the right bank of the River Volga 2.5 km downstream from

Table 1. The range of variations in J_n and K_m in the studied rocks

No.	Age of rock/outcrop	Number of samples	Range of variations in J_n, mA/m	Mean J_n, mA/m	Number of samples	Range of variations in K_m, 10^{-6} SI units	Mean K_m, 10^{-6} SI units
1	C_3g, Gzhel site	11	6–47	23.0	11	164–286	215
2	C_2m, Rzhev	18	0.4–15.2	4.1	12	43–441	119
3	C_2m, Nara River	26	0.7–31.6	7.2	7	115–309	177
4	C_2b, River Volonga	59	0.01–14.4	1.4			
5	C_1v, Boksitogorsk	57	0.6–157.0	27.1	57	57–1131	257

the Rzhev suburbs. The samples represent brick-red clays from the 15-m interval of the sequence containing 20 stratigraphic levels (*Geologiya*..., 1971). The bedding is close to horizontal, the average coordinates of sampling are 56.23° N, 34.41° E, Fig. 1.

North Timan, River Volonga (Malaya Pokayama stow) (collection of G.I. Goncharov, 1969). The Ivarskaya and Yalumskaya groups of the Bahkirian stage of the Carboniferous. The samples represent gray, red, and cream-colored marly dolomites. Thirty-six meters of the sequence and 17 stratigraphic levels are sampled (*Opornye*..., 1997). The bedding varies within 150°–196° in the dip azimuth and within 25°–47° in the dip angle; the sampling coordinates are 67.08° N, 48.04° E, Fig. 1.

Leningrad Oblast. The Carboniferous rocks of the Tula horizon of the Visean stage (collection of Khramov, 1961) are sampled at a distance of 4 km from the town of Boksitogorsk in the southern and western wall of the quarry of mine no. 13. The rock samples selected for the analysis are represented by lilac brown, brown, and brownish red clays, and lilac brown and brick-red bauxites (total thickness 40 m, 3 stratigraphic levels) (*Geologicheskaya*..., 1968; *Geologiya*..., 1971). The bedding is close to horizontal, the average coordinates of sampling 59.48° N, 33.89° E, Fig. 1.

The laboratory paleomagnetic studies and processing of the obtained results were conducted by the standard procedure (*Paleomagnitologiya*..., 1982; Khramov, 1987). Thermal demagnetization of the samples was conducted in the facilities placed into five-layer permalloy screens (VNIGRI, original design) and TD48 (United States). The NRM was measured by magnetometers JR-4 and JR-5 (AGICO, Czech Republic). Magnetic susceptibility was measured by the KLY-3 and MFK1-FA kappa-bridges (AGICO, Czech Republic). For analyzing and visualizing the obtained results, we used the program packages (Popov, 2007; 2008; Enkin, 1994; Chadima and Jelinek, 2009). The fold tests were used in the modification (Watson and Enkin, 1993). The trajectories of the apparent polar wander path (APWP) were calculated by the GMAP program (Tarsvik et al., 1990). In the data processing, the statistics was taken at the level of the rock pieces (stratigraphic levels).

EXPERIMENTAL DATA

Magnetic properties. The scalar magnetic characteristics—natural remanent magnetization (J_n) and magnetic susceptibility (K_m) for the studied rocks vary widely (Table 1). The heating and cooling curves of K_m for Gzhel clays (Fig. 2, samples 6419 and 6423) show that J_n is mainly carried by magnetite and hematite. The characteristic peak in the cooling curves in the interval 150–200°C is probably the result of the release of the stress state associated with maghemitization. This is consistent with the data of O.L. Bagina about the composition of the J_n-carrier magnetic minerals of Gzhelian clays at the Gzhel site (Bagina, 1966; 1968).

For the rocks from the outcrop in the vicinity of Boksitogorsk, the behavior of K_m during heating and cooling testifies to the presence of magnetite and hematite in some samples and hematite alone in other samples (Fig. 2, samples 946 and 947). In both collections, the samples are strongly chemically altered and the cooling curves show the formation of magnetite, probably due to the disintegration of the iron-bearing nonmagnetic minerals (Fig. 2).

The distributions of the directions of the principal axes for the magnetic susceptibility anisotropy ellipsoids in the Gzhelian (Gzhel site) and Visean rocks (Boksitogorsk) are similar to the characteristic distributions of the sediments (Figs. 3a and 3b). The anisotropy ratio of the magnetic particles (P = K1/K3) is at most 10%. The average anisotropy ratio is P = 1.07 and P = 1.03 for the Gzhelian and Visean clays, respectively. The Gzhelian clays are dominated by planar anisotropy, whereas in the Visean clays from Boksitogorsk, there are more particles with a low anisotropy ratio. This may indicate that the rocks of the Boksitogorsk sequences are not abundant in large particles and aggregates of small particles.

The petromagnetic characteristics of the Carboniferous rocks of River Volonga were studied by N.V. Lozhkina et al. (*Opornye*..., 1997). The presence of iron sulfides and hematite is noted. In our studies, the

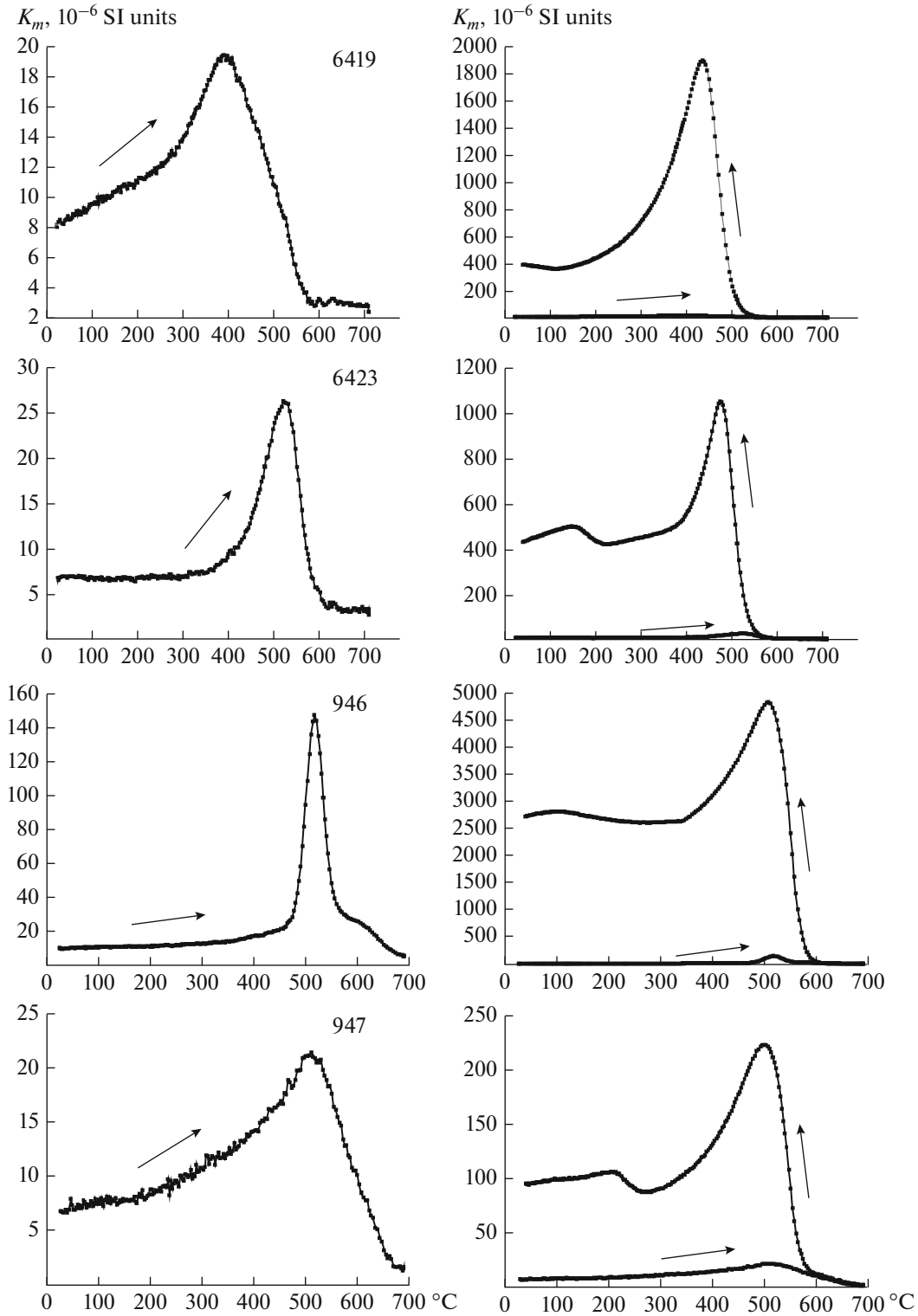


Fig. 2. The examples of thermomagnetic analysis of the Carboniferous rocks: the curves of the behavior of magnetic susceptibility at heating (right arrow) and cooling (left arrow). Samples 6419, 6423 are Gzhelian clays (Gzhel site); samples 946, 947 are Visean clays (Boksitogorsk).

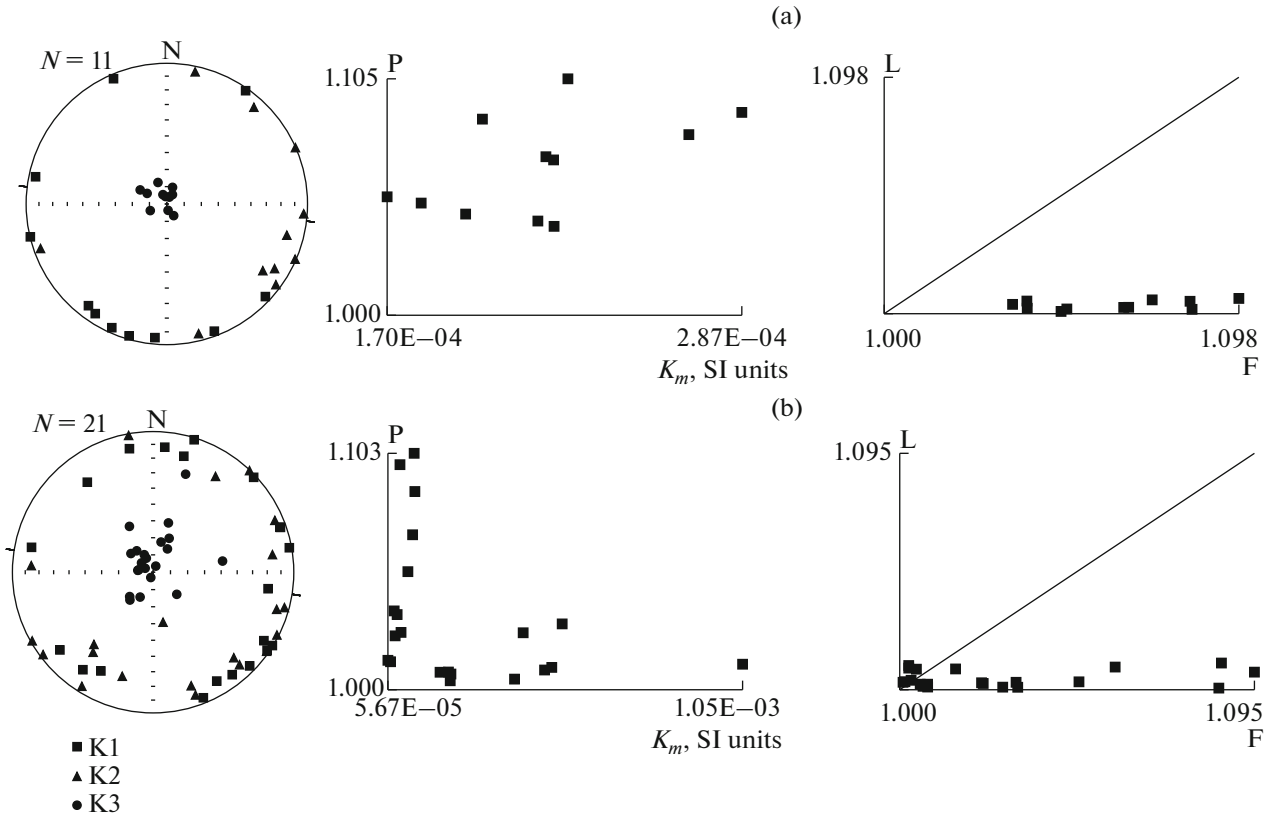


Fig. 3. The distributions of the directions of the principal axes of the magnetic susceptibility ellipsoids: (a) Gzhelian clays; (b) Visean clays; K1, K2, and K3 are the maximal, intermediate, and minimal semiaxes of the magnetic susceptibility anisotropy ellipsoid; P is anisotropy ratio; K_m is magnetic susceptibility, F is the oblateness, and L is elongation of the ellipsoid.

deblocking temperatures of the isolated J_n components in the rock samples from River Volonga vary within 200–500 and 200–640°C. The minerals of the magnetite series and hematite are the main carriers of magnetization in the studied rocks.

Component analysis of J_n. The J_n components were isolated by stepwise thermal demagnetization. The J_n component in the low-temperature interval (100–250°C) was not identified. Since all the samples were stored in a laboratory in arbitrary positions for many years, the information about the low-temperature (viscous) J_n component has been destroyed. However, this component is insignificant in most red rocks. In the middle and high temperature intervals of thermal demagnetization, as seen from the Zijderveld diagram (Fig. 4), there are samples with one-component and two-component J_n.

Two J_n components are revealed in the Gzhelian rocks. The first (midtemperature) component, C3, is isolated in the temperature interval 200–570°C, and the second (high-temperature) component, C0, in the interval 570–670°C (Fig. 4, samples 6418V and 6424N). One sample (6423N) in which the first J_n component is isolated in the temperature interval 200–640°C is an exception. The distributions of components C3 and C0 for the rocks of the Gzhelian age are illustrated in Figs. 5a and 5b, and their statistical characteristics are presented in Table 2.

The high-temperature components C2 and C0 in the rocks from the River Volga outcrop (Rzhev) are identified in the temperature intervals 200–630 and 575–675°C, respectively (Fig. 4, sample N9711). Component C0 is absent in the part of the samples, whereas component C2 in these samples is isolated in the interval 200–675°C (Fig. 4, N972). The distribution of components C2 and C0 for the rocks from the outcrop of River Volga is illustrated in Figs. 5c and 5d and their statistical characteristics are presented in Table 2.

The NRM of the Moscovian stage Vereya rocks in the region of Nara River has a single component (Fig. 4, samples N973 and N978). The high-temperature component C2 is isolated in the temperature interval 200–675°C. The distribution of component C2 for the rocks from the outcrop on the Nara River is shown in Fig. 5e and its statistical characteristics are indicated in Table 2.

The samples from the outcrop in the basin of River Volonga (northern Timan) were split into two groups during thermal demagnetization. One group (the Yal-

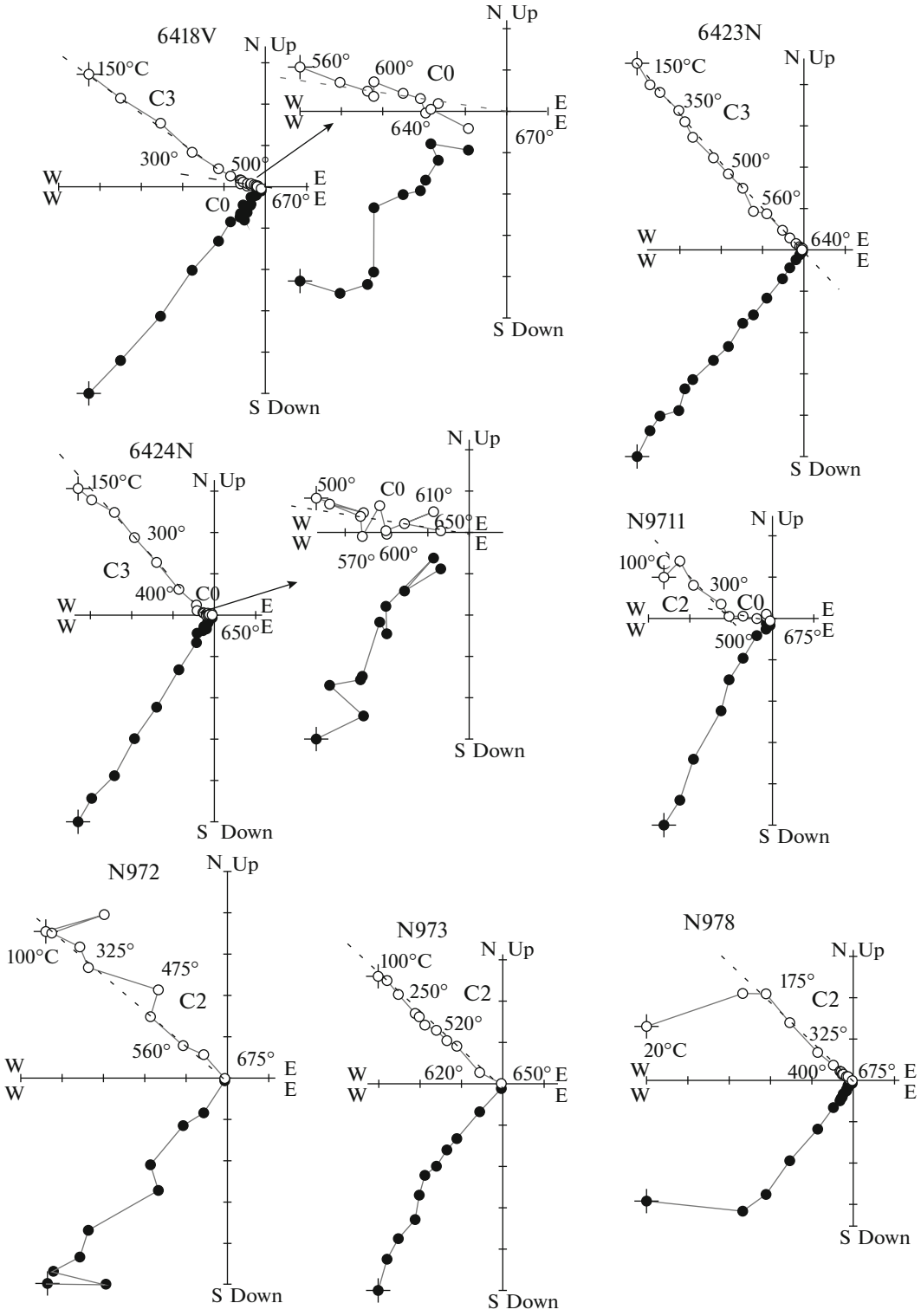


Fig. 4. The examples of stepwise thermal demagnetization of the Carboniferous samples. Samples 6418V, 6423N, 6424N represent the Gzhelian horizon of the Moscovian age Carboniferous rocks; samples N9711 and N972 represent the Kashirian horizon of the Moscovian age; samples N973 and N978 represent the Virean horizon of the Moscovian age; samples 964-4, 967-1, and 968-1 represent the Bashkirian and samples 947N, B-956, N-962, 959B, the Visean Carboniferous rocks. The Zijderveld diagram is shown in the stratigraphic coordinates; the open and filled circles are vector projections on the horizontal and vertical plane, respectively. The dashed line shows the intervals of isolation of the J_n components (their names are indicated in accordance with Table 2). The Zijderveld diagrams are constructed with the use of Remasoft 3.0 program (Chadima and Hrounda, 2006).

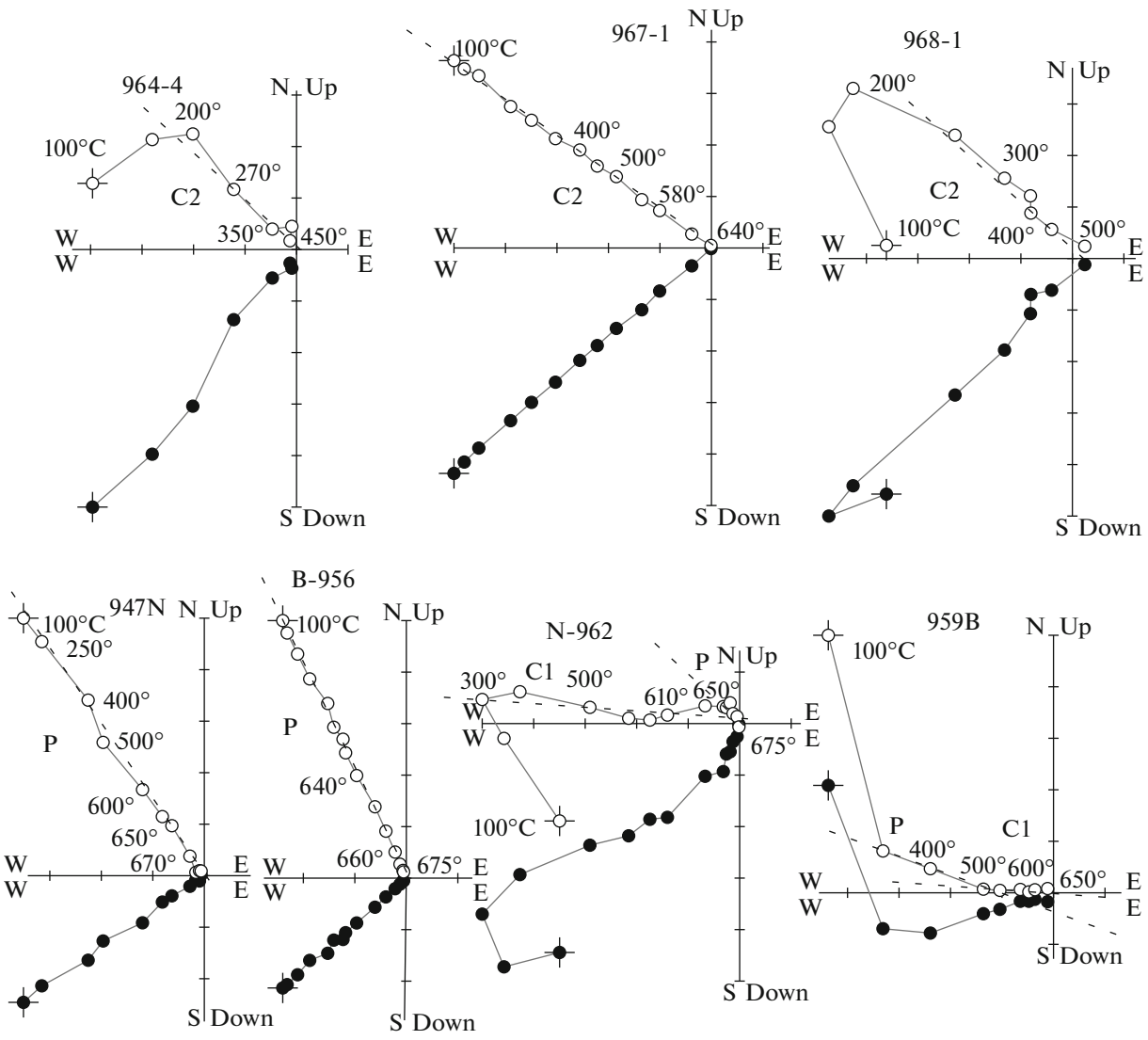


Fig. 4. (Contd.).

umskaya series gray limy dolomites) demonstrates the chaotic variations of the J_n direction even after heating to 150–200°C and has a determination error of above 20–50°C. These samples make up about 40% of the collection. In the remaining samples (red and creamy-colored marly dolomites of the Ivarskaya series), the C2 component is isolated in the temperature intervals 200–500 and 200–650°C (Fig. 4, samples 964-4, 967-1, and 968-1). The fold test (Watson and Enkin, 1993) for the C2 component is positive F+, (0.6 ± 0.4). The 64% unfolding test (Watson and Enkin, 1993) gives the following J_n direction: $N = 12$ D = 217, J = –23, K = 158, and $\alpha_{95} = 4°$ (Fig. 6). The distribution of component C2 for the rock samples from the region of River Volonga is shown in Fig. 5f and the statistical characteristics are presented in Table 2. The studies of the Bashkirean age rocks from the Volonga river outcrops conducted by N.V. Lozhkina et al. have not identified the statistically significant J_n direction (*Opornye...*, 1997). The selective thermal demagnetization of the Bashkirian sediments of River Volonga 250–300° during their initial investigation, according to the oral communication of Goncharov, revealed the J_n direction with N = 11, D = 223, J = –21, K = 17, and $\alpha_{95} = 11°$. The stepwise thermal demagnetization of the samples from the outcrop of the Volonga river shows that the C2 component appears in the temperature interval 200–300°C (Fig. 4, samples 964-4, 967-1, 968-1). This allows us to unite the data of Goncharov with the results obtained in our study (Table 3, lines 11–13).

The Visean rocks (Boksitogorsk) have two J_n components. The first component, P, is identified in some samples (Fig. 4, samples 947N, B-956) in the temperature interval 200–675°C, and in the other samples in the interval 630–675°C (Fig. 4, sample N-962).

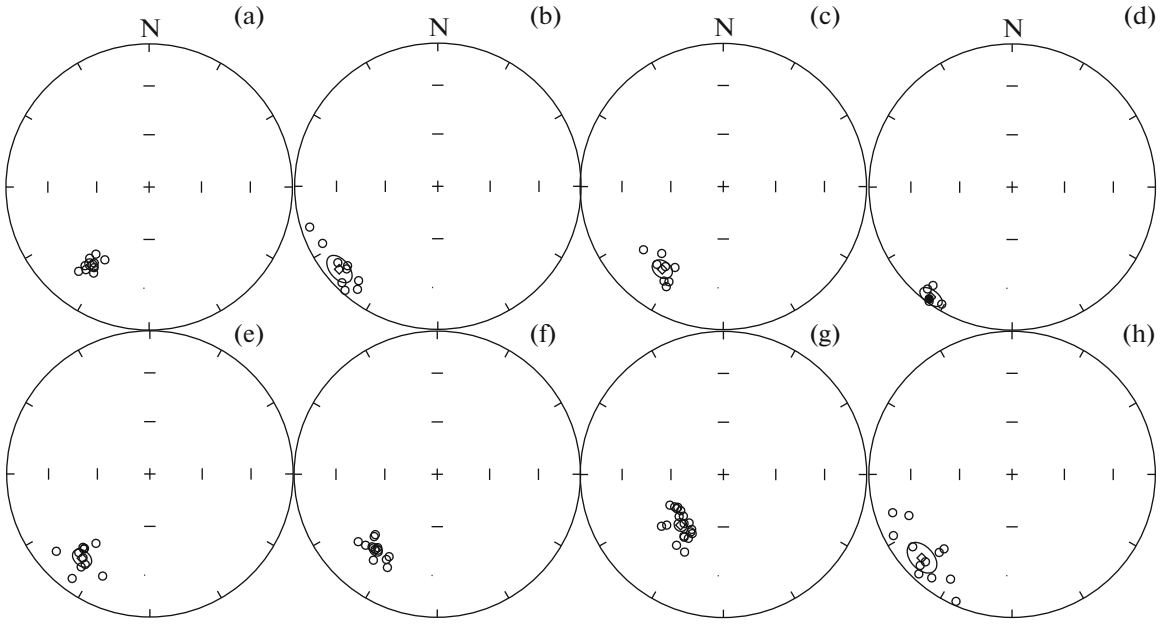


Fig. 5. The distributions of the directions of NRM components after demagnetizations: (a), (b), Gzhelian rocks, Gzhel site; (c), (d), Moscovian rocks, Rzhev; (e), Moscovian rocks, Nara river valley; (f), Bashkirian rocks, River Volonga, North Timan; (g), (h), Visean rocks, Boksitogorsk.. The open and filled circles are vector projections on the upper and lower hemispheres, respectively. The distributions are shown in the stratigraphic coordinate system.

The second component, C1, is also isolated in two different temperature intervals, 300–620 and 640–670°C (Fig. 4, samples N-962 and 959B, respectively). The distributions of components P and C1 for the Visean rocks is shown in Figs. 5g and 5h, and their statistical characteristics are presented in Table 2. The P component in the Boksitogorsk Visean rocks has very good convergence with Khramov's data for these rocks (*Paleomagnitnye*..., 1979), Table 3. For comparison, we present the paleomagnetic pole for the Visean rocks obtained by R.A. Komissarova from the Azov limestones for the C1 component (*Paleomagnetic*..., 1971). For comparison, in Table 3 we present the paleomagnetic pole for the Late Devonian-Early Carboniferous rocks of the Vaigach Island, corrected for tectonic rotation (Zhuravlev et al., 2011; Zhuravlev and Iosifidi, 2012).

Based on the obtained determinations and the data contained in the paleomagnetic database (McElhinny and Lock, 1996; Pisarevsky, 2005), the trajectory of the APWP was constructed in the Devonian–Early Permian interval by the spherical spline method (Jupp and Kent, 1987). The paleomagnetic data from the Devonian, Carboniferous, and Early Permian rocks of the Russian Platform and Central and Western Europe were used (Table 4). In Table 4 we separately present the paleomagnetic poles from part of the British Isles (Scotland and England), which are used by different authors for constructing the APWP for the East European Platform. The constructed APWP curve for VEP (Fig. 7) permitted estimating the kinematical parameters of VEP motion in the Carboniferous (Table 5).

DISCUSSION

The obtained paleomagnetic directions from the Carboniferous rocks of the Gzhel site, River Volga, and Nara River have good convergence in components C3 and C2 not only with the results of the first researchers of the studied collection (Table 3) but also with the paleomagnetic determinations from the coeval rocks of the Donbass sequences (Table 4, lines 16–18 and 23–24). For the rocks of the Gzhelian age, the average paleomagnetic pole over the data of the previous researchers (Table 3, lines 1–4) is $N = 4$, $\Phi = 43°$ N, $\Lambda = 170°$ E, K = 393, and $\alpha_{95} = 4°$). The total average test which is similar to the reversal test (McFadden and McElhinny, 1990) for this pole and the pole obtained in the present work (Table 3, line 5) is positive and corresponds to class B. The angle between the average poles is 2° and the critical angle is 6° (McFadden and McElhinny, 1990). This allows us to unite the data for the Gzhelian stage (Table 3, line 6). The paleomagnetic poles from the C0 components have positions that are close to the poles of the Early Carboniferous, which is inconsistent with the determinations for the Early Carboniferous presented in the World paleomagnetic database (McElhinny and Lock, 1996; Pisarevsky, 2005). The isolated C0 components (deblocking temperature interval 575–675°C) that have low inclinations are most likely to be associated with the large hematite particles and particle aggregates created due to the interaction of the fine magnetic grains with clay particles. The studies of the different rock types of the Late Carboniferous of the

Table 2. Paleomagnetic directions and poles for the studied Carboniferous rocks

No.	Age	J_n component (isolation interval, °C)	c.s.	N/n	D°	I°	K	α_{95}°	Φ, °N	Λ, °E	dp°	dm°	φ_m°
Moscow region, Gzhel site, Shchelkovo clays, Gzhelian Stage													
1	C_3g	**C3** (200–570°C)	s	9/9	217	–33	182	4	43	167	3	5	18
2		**C0** (570–670°C)	s	9/9	230	–11	44	8	26	160	4	8	6
Tver region, Rzhev, Rzhev layers, Kashirian horizon, Moscovian Stage													
3	C_2m	**C2** (200–630°C)	c	8/14	216	–29	84	6	41	166	4	7	16
4		**C0** (575–675°C)	s	5/7	216	–5	135	7	29	172	4	7	3
Pushchino, right bank of River Nara, Vereyan horizon, Moscow Stage													
5	C_2m	**C2** (200–675°C)	s	11/20	219	–25	69	6	38	166	4	6	13
River Volonga (Malaya Pokayama), Ivarskaya and Yalumskaya series, Bashkirian Stage													
6	C_2b	**C2** (200–500°C, 200–650°C)	c	12/24	215	–5	104	4					
			s		219	–32	137	4	34	180	3	5	17
7		Fold test, 64% unfolding	s	12/24	217	–23	158	4	30	185	2	4	12
Leningrad region, Boksitogorsk, Tula horizon, Visean Stage													
8	C_1v	**P** (200–670°C, 600–670°C)	s	19/30	220	–52	80	4	52	151	4	6	33
9		**C1** (200–600°C, 640–670°C)	s	12/23	227	–15	25	9	27	159	5	9	8

c.s. is the coordinate system (g, geographic; s, stratigraphic); N, number of samples; n, number of specimens; D°, I° declination and inclination of the mean directions of the magnetization components; K, the precision parameter of the vectors; α_{95}°, the radius of the confidence circle at a 95% probability for the mean direction; Φ°, Λ°, the latitude and longitude of the paleomagnetic pole; dp°, dm°, the semi-axis of the confidence oval; and φ_m°, the paleomagnetic latitude.

Donbass have demonstrated the presence of the J_n component with a low inclination, which is associated with the presence of large hematite grains (Iosifidi and Khramov, 2002; Iosifidi et al., 2010). Similar conclusions based on the first studies of the Gzhekian clays were made by Bagina in (Bagina, 1966; 1968). The study of the red Gzhelian rocks of Donbass (Iosifidi et al., 2010) has also revealed an underestimation of the inclination for the high-temperature component J_n which is, in the opinion of the authors, due to the presence of the large hematite grains.

The direction of the high-temperature J_n component in the Bashkirian rocks of River Volonga is prefolding; the fold test is positive (Table 2, Fig. 7). The 64% unfolding gives the paleomagnetic pole Φ = 30° N, Λ = 185° E, and α_{95} = 3°, which is closer to the more ancient Carboniferous paleomagnetic poles of the Russian Platform. We note that the secondary remagnetization of the rocks during folding is not necessarily the only cause of the formation of the synfolding component. The rocks of the Bashkirian age in River Volonga (Ivarskaya series) transgressively, with an angular inconformity overlie the dolomite breccia of the Suvoinaya series (Lower Carboniferous) within which thrusting of the layers is clearly seen (*Opornye*..., 1997). The upper boundary of the Suvoinaya series is gently undulating with the erosional downcuts up to 3 m (*Opornye*..., 1997).

The presence of subaeral unconformity between the Ivarskaya and Suvoinaya rocks (Artyushkov and Chekhovich, 2011) suggests that the synfolding component is associated with the primary inclination of the plane of the ancient horizon for the rocks of the Ivarskaya series (the average inclination is a few degrees). The difference (in longitude) of the paleomagnetic polar direction for the rocks of the Bashkirian stage of River Volonga is due to the fact that the Mid-Paleozoic structures of the North Timan have probably experienced counterclockwise rotations under the tectonic activation during the EEP collision with the Urals structures due to thrusting (*Geodynamicheskaya*..., 1988; Iosifidi et al., 2007). The correction of the obtained paleomagnetic pole for the tectonic rotation will be possible after obtaining the key paleomagnetic pole for the Bashkirian time from the deposits of the Russian Platform.

For the Tula horizon rocks of the Visean stage from Boksitogirsk there is a close consistency of the data for the P-component with the data of the first study (Table 3, lines 14–16). The mean paleomagnetic pole is close to that for the Permian rocks of the Russian Platform and testifies to the secondary origin of component P. Component C1 identified in the Visean rocks (Bositogorsk), in contrast to the C3 and C2 components of the Gzhelian and Moscovian stages (Gzhel site and River Volga) has the same direction in two temperature intervals 300–620 and 640–670°C.

%	D°	I°	K	α_{95}°
−50	215	9	62	6
−40	215	7	69	5
−30	215	4	76	5
−20	215	1	85	5
−10	215	−2	94	5
0	215	−5	104	4
10	215	−8	115	4
20	216	−10	126	4
30	216	−13	137	4
40	216	−16	147	4
50	216	−19	154	4
60	217	−21	158	4
70	217	−24	158	4
80	218	−27	153	4
90	218	−30	144	4
100	219	−32	132	4
110	220	−35	119	4
120	221	−38	105	4
130	222	−40	92	5
140	223	−43	80	5
150	224	−46	69	5

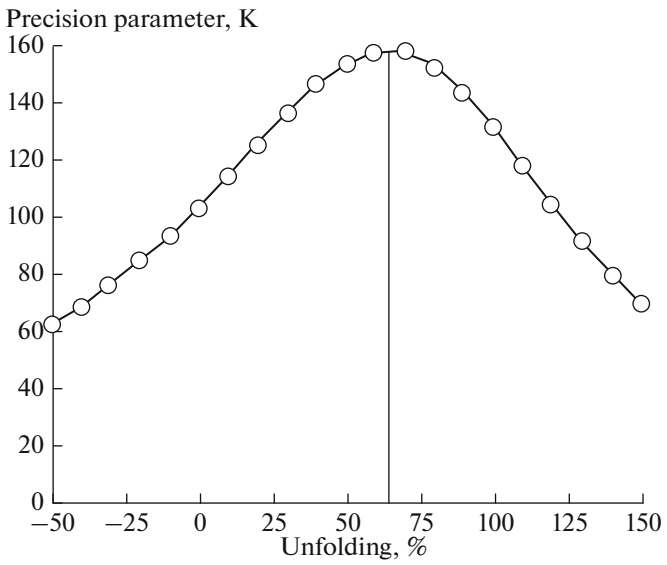


Fig. 6. The fold test for the North Timan data. % is the percentage of unfolding; D°, I° are the declination and inclination of the mean directions of magnetization components; K is the precision parameter of the vectors; α_{95}° is the radius of the confidence circle at a 95% probability for the mean direction.

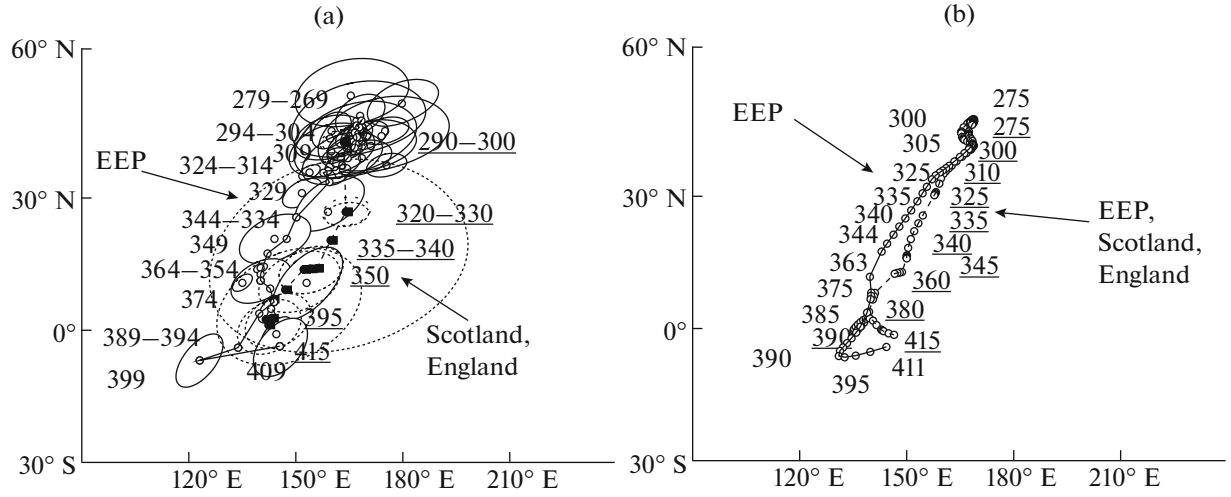


Fig. 7. The APWP curve in the interval 270–410 Ma for the EEP. (a) The trajectory of the migration of the paleomagnetic pole for the EEP (solid line) and the British Isles (dashed line) (according to the data of Table 4, moving average); (b) the APWP curve (spline) for the EEP (solid line) and the British Isles (dashed line); the age of the points for the EEP and the British Isles is indicated left and right (underscored) of the curve, respectively. The confidence ellipses for the individual points are shown by the solid line for the EEP and by the dashed line for the British Isles. The plot is drawn with the use of GMAP program (Torsvik et al., 1990).

This indicates that this component is carried by the magnetic minerals of the magnetite and hematite series. The predominance of particles with a low anisotropy ratio in the Visean clays (Boksitogorsk) testifies to the absence of the large particles/particle aggregates which give underestimated values of the inclination for the high-temperature components of the Gzhelian clays, and the C1 component, to all appearances, has a detrital and chemical origin. At the present stage of studying the Carboniferous rocks of

Table 3. The comparison of the obtained paleomagnetic poles with the results of the first studies of the considered collections for the Carboniferous rocks of the Russian Platform

No.	Index in the World Paleomagnetic Database (Result no)	φ, °N	λ, °E	Φ, ° N	Λ, °E	dp°	dm°	Age, Ma
Gzhelian Stage								
1	4901	56	38.5	42	169	1	2	300
2	4902	55	39	41	167	3	4	301
3	4900	55.6	38.5	42	167	1	2	301
4	4899	55.6	38.5	46	177	3	5	302
5	t. w.	55.6	38.5	43	167	3	5	302
6		C_3g	$N = 5$	43	169	$\alpha_{95} = 3°$		302
Moscovian Stage								
7	t. w.	56.2	34.4	41	166	4	7	312
8	4913	56.5	34.5	33	159	2	4	313
9	t. w.	54.9	37.4	38	166	4	6	315
10		C_2m	$N = 3$	37	164	$\alpha_{95} = 5°$		313
Bashkirian Stage, North Timan								
11	t.w.	67.08	48.04	30	185	2	4	
12*		67.08	48.04	27	179	6	12	
13**		C_2b	$N = 2$	28	182	$\alpha_{95} = 4°$		322
Visean Stage, P component								
14	t.w.	59.5	33.9	52	151	4	6	
15	4938	59.5	34	49	164	6	4	
16**		C_1–C_2	$N = 2$	51	158	$\alpha_{95} = 4°$		265–270
Visean Stage, C2 component								
17	4933	48	38	27	167	6	12	340
18	t. w.	59.5	33.9	27	159	5	9	335–345
Late Devonian–Early Carboniferous rocks of Vaigach Island, D3–C1								
19***		70.4	58.8	19	158	4	8	350

φ, °N; λ, °E, the coordinates of sampling; Φ°, Λ°, the latitude and longitude of the paleomagnetic pole; dp°, dm°, the axes of the confidence oval; α°_{95}, the radius of the confidence circle at a 95% probability for the direction of the pole; the indices of the results correspond to the Resultno of the World's Paleomagnetic Database (WPDB) (McElhinny and Lock, 1996; Pisarevsky, 2005); t.w. stands for this work; the sign * indicates an oral report of Goncharov; the sign ** indicates that data averaging over the R-vector—resultant of NRM; sign *** denotes the works (Zhuravlev et al., 2011; Zhuravlev and Iosifidi, 2012).

the Russian Platform, we assume that component C1 is ancient.

All the isolated J_n components have in common the correspondence of their directions to the geomagnetic field of reversed polarity. Due to the closeness of the Kiama hyperchron, this can point to the Late Paleozoic remagnetization of the rocks. The correlation of the obtained paleomagnetic poles to the existing data from the EEP (Table 4) suggests that the rocks were remagnetized only partially and that the remagnetization process was close to the formation time of these sediments. The closeness of the acquisition time of secondary (chemical) magnetization and the formation time of the rocks is, inter alia, supported by the isolation of three identical J_n components in the three different magnetic minerals—primary magnetite, maghemite, and hematite for Gzhelian rocks (Fig. 4). This was noted even for the first studies of the Gzhelian clays by O.L. Bagina (Bagina, 1966; 1968). The isolation of the C0 and C1 components (Table 2) shows the absence of the complete Late Paleozoic remagnetization and supports the closeness in time between the emergence of chemical magnetization and the formation of the rocks.

For estimating the kinematical parameters of the EEP in the Carboniferous from the existing paleomagnetic data (Table 4), we constructed the APWP curves for the EEP and separately for the data for the British

Table 4. The paleomagnetic poles and the coordinates of the APWP curve for the EEP in the Devonian–Early Permian

No.	Index in WPDB (Result no)/references	Rock type	Sampling coordinates		Q	Coordinates of the pole		dp°	dm°	APWP		Age, Ma
			φ, °N	λ, °E		Φ, °N	Λ, °E			Φ, °N	Λ, °E	
1	2	3	4	5	6	7	8	9	10	11	12	13
1	4816	Sediments, red rock	56.0	53.0	4	43.0	167.0	2.0	3.0	45.3	167.8	269.0
2	/1	Sediments, red rocks	57.7	55.6	4	47.0	168.0	3.0	5.0	45.4	168.0	270.0
3	7342	Terrigenous sediments	54.8	56.7	3	44.0	160.0	3.0	4.0	45.7	168.8	280.0
4	/2	Trachites	47.6	37.8	5	49.4	179.7	4.0	7.0	45.4	168.6	282.6
5	36	Dolerites	58.6	11.3	3	50.9	165.4	6.0	10.2			285.0
6	3241	Sediments, red rocks	49.9	15.6	5	40.5	164.3	3.5	3.5			285.0
7	6311	Terrigenous sediments	50.6	45.5	4	40.0	166.0	1.0	2.0			285.0
8	3620	Igneous rocks	50.0	8.0	3	46.0	167.0	7.0	13.0	43.4	166.7	291.0
9	6314	Granites	51.0	16.0	6	42.0	166.0	7.0	13.0	43.0	166.6	293.0
10	8937	Sediments, red rocks	48.5	37.8	6	43.0	164.0	2.0	3.0			295.0
11	1467	Sediments, igneous rocks	50.7	10.9	4	41.0	171.0	5.0	9.0	42.6	166.8	295.0
12	/3	Limestones, sandstones, red rocks	48.5	37.8	6	43.0	165.0	2.0	3.0	42.2	167.2	297.0
13	2868	Diabase dikes	58.4	8.8	3	42.5	159.6	4.3	7.7	41.9	167.6	297.0
14	3047	Diabases	50.0	7.8	4	39.0	161.0	2.0	3.0			297.0
15	5823	Dolerites	55.5	13.5	4	38.5	168.5	3.4	6.7	41.6	168.1	300.0
16	t. w.	Sediments, red rocks	57.0	35.0	5	43.0	169.0	3.0	5.0	41.1	168.3	302.0
17	/3	Limestones, sandstones, red rocks	48.5	37.8	6	42.0	179.0	2.0	4.0	40.9	168.3	302.5
18	/4	Limestones, sandstones, red rocks	48.3	38.3	5	44.0	167.0	1.0	2.0	40.8	168.2	303.0
19	3169	Igneous rocks	50.5	16.5	3	43.0	174.0	7.0	13.8			303.0
20	3777	Tuffs, sandstones	50.6	16.1	4	37.0	175.4	2.3	4.6			304.5
21	3240	Sediments, red rocks	50.0	15.0	5	38.4	162.7	8.2	8.2	39.9	167.4	305.0
22	3333	Sediments, red rocks	50.2	14.0	4	35.5	153.8	1.3	2.6	38.4	165.1	308.5
23	t.w.	Sediments, red rocks	57.0	35.0	6	37.0	164.0	4.0	4.0	36.6	162.3	313.0
24	/4	Limestones, sandstones, red rocks	48.8	38.6	5	35.0	162.0	2.0	3.0	36.1	161.4	314.5
25	3332	Sediments, red rocks	49.8	13.3	4	31.1	151.7	2.6	5.1	35.1	159.6	318.0
26	5814	Sediments	59.7	10.4	4	39.9	159.9	5.0	9.4	32.1	155.7	329.0
27	t. w.	Sediments, red rocks	57.0	35.0	4	27.0	159.0	5.0	9.0	27.0	151.3	335.0

Table 4. (Contd.)

No.	Index in WPDB (Result no)/references	Rock type	Sampling coordinates		Q	Coordinates of the pole		dp°	dm°	APWP		Age, Ma
			φ, °N	λ, °E		Φ, °N	Λ, °E			Φ, °N	Λ, °E	
1	2	3	4	5	6	7	8	9	10	11	12	13
28	5818	Sediments, red rocks	61.4	5.5	4	21.0	144.0	5.0	10.0	21.6	146.3	340.0
29	8961	Dolerites	71.0	25.7	5	14.4	140.1	4.4	8.6	17.7	142.9	344.0
30	5819	Lamprophyres	63.1	11.0	4	11.5	140.1	4.5	8.7	12.0	139.7	363.0
31	/5, 6	Sediments, red rocks	59.9	32.7	6	11.0	135.0	2.0	3.0	8.3	140.1	375.0
32	6759	Andesites	47.5	38.0	3	11.0	153.0	5.8	11.3			375.0
33	6855	Tuffs, sandstones	47.6	37.6	5	5.0	143.0	2.1	4.2	6.8	139.8	377.0
34	6856	Basalts, tuffs, spillites	47.6	37.6	6	−0.9	144.5	6.0	11.0	0.4	137.0	384.0
35	6866	Limestones, sandstones, red rocks	47.6	37.6	6	−7.0	123.0	7.0	14.0	−5.2	131.3	392.0
36	6958	Sediments, red rocks	48.7	26.0	6	−3.7	145.5	5.2	8.7	−4.3	144.3	411.0
		Data for the British Isles										
37	9223	Dolerites	55.2	358.0	5	47.1	157.1	7.7	7.7			290.0
38	6321	Dolerites	54.8	355.0	4	45.9	162.7	3.8	7.2	43.4	166.7	291.0
39	3008	Dolerites	55.5	358.3	4	44.4	159.1	2.7	5.1			294.0
40	9221	Dolerites	55.7	352.2	5	35.4	166.8	3.2	6.3			294.0
41	6316	Quartz dolerites	56.2	355.0	4	38.3	174.0	3.0	5.0			294.5
42	7473	Dolerites	54.0	357.8	5	39.4	161.6	2.4	4.7			294.5
43	865	Basic dikes	57.5	358.2	3	41.3	161.8	0.7	1.3	42.6	166.8	294.5
44	6305	Igneous rocks	56.0	357.5	4	27.0	164.0	3.0	6.0	30.7	154.5	330.0
45	6299	Basalts	53.2	358.2	6	14.3	155.9	5.5	9.5	17.7	142.9	345.0
46	6315	Basalts	56.2	356.9	6	14.1	152.2	4.0	8.0	12.0	139.7	359.0
47	1577	Sediments, red rocks	58.5	356.8	5	9.4	147.0	4.5	7.9	−6.7	132.6	400.0
48	2531	Igneous rocks	57.0	357.0	6	5.0	140.0	5.0	9.0	−6.3	135.7	405.0
49	866	Basalts, andesites	56.5	357.0	7	−2.2	138.2	2.7	4.3	−5.4	139.7	410.0
50	2582	Igneous rocks	56.5	354.5	4	−2.0	141.0	5.0	8.0	−4.3	144.3	410.0
51	6615	Ignimbrites	56.6	357.2	5	4.0	150.0	2.0	3.0			415.0

φ, °N; λ, °E, the sampling coordinates; Φ°, Λ°, the latitude and longitude of paleomagnetic pole; dp°, dm°, the semiaxes of the confidence oval; Q, reliability criterion of paleomagnetic data (Van der Voo, 1990); references: 1, (Iosifidi and Popov, 2015); 2, (Yuan et al., 2011); 3, (Iosifidi et al., 2010); 4, (Meijers et al., 2010); 5, (Gurevich et al., 2005); 6, (Rodionov et al., 2010); t. w., this work.

Table 5. The kinematic parameters of the EEP in the Carboniferous

Interval, Ma	Rotation angle, $\Delta\psi^o$	Rotation pole		Latitudinal displacement of the plate, $\Delta\varphi^o$	Plate velocity, cm/yr, V_φ	Plate velocity deg/Ma	
		Φ, °N	Λ, °E			V_φ	V_ψ
299–309	4.0	21.5	56.9	2.4	2.7	0.2	0.4
309–319	6.0	41.0	31.0	0.8	0.9	0.1	0.6
319–329	4.0	35.0	40.0	1.3	1.4	0.1	0.4
329–339	12.7	32.7	42.0	3.9	4.3	0.4	1.3
339–349	7.6	35.8	40.0	3.7	4.1	0.4	0.8
349–359	1.6	23.3	44.9	0.9	1.0	0.1	0.2

$\Delta\psi^\circ$ is the angle of plate rotation; $\Delta\varphi^\circ$ is the latitudinal displacement of the plate; V_ψ is plate rotation angle; V_φ is the velocity of the latitudinal displacement of the plate; latitudinal displacements are calculated for the EEP center 57° N, 35° E.

Isles (Scotland and England) (Fig. 7). When picking the data for constructing the APWP curve, we preferred the paleomagnetic determinations obtained from the sedimentary rocks (67%). The data for the British Isles are mainly represented by the determinations from igneous rocks (90%). We note that there are half as many reliable data for the Early Carboniferous than for the Late Carboniferous. The APWP curves for the EEP and from the British Isles (Scotland and England) obtained by the moving average (with a 20-Ma window shifted by 5 Ma) (Fig. 7a) indicates the eastward shift of the APWP curves for the British Isles and the more northern positions of the ancient paleomagnetic poles. These distinctions, at first glance, fall within the APWP construction errors; however, the spline trajectories clearly fix an appreciable shift of the data from Devonian, which indicates the presence of shear deformations. The eastward displacement of the Carboniferous poles is likely to be associated with the manifestation of thrusting processes (Fig. 7b). Hence, when the data for Scotland and England are used, the correction for the tectonics that took place in the Hercynian time during the assembly of the Pangaea should be introduced. The influence of the tectonics of that period on the paleomagnetic data for the Carboniferous rocks of Central Europe (South Poland) is revealed, e.g., in (Nawrocki et al., 2008).

Based on the APWP curve constructed for the EEP, we estimated the kinematic parameters of this platform during the Carboniferous (Table 5). The total rotation angle is 36° ± 6° and the lateral displacement is 13° ± 7°. The relatively high EEP rotation velocity falls in the interval 349–329 Ma (Table 5). The velocity of the lateral drift is rather low. The maximal values fall in the same time interval 349–329 Ma (Table 5).

The obtained estimates of the kinematical parameters of the EEP in the Carboniferous (359–299 Ma) differ from the data of the other works. For instance, based on the extensive paleomagnetic data for the British Isles (Devonian–Carboniferous), Torsvik et al. (2012) estimate the total rotation angle at 45° with the average velocity 0.75 degrees per Ma (the rotation pole is at φ = 37.5° N, λ = 52.0° E); according to these authors, the latitudinal shift is 31° with the average velocity of 0.52 degree per annum (5.7 cm per annum). The source of these discrepancies primarily is the incompleteness of the paleomagnetic time series for the Early Carboniferous for both the Russian Platform and the western part of the EEP. This necessitates using the data for the neighboring tectonic blocks which, as demonstrated by the example of the data for North Timan and the British Isles, require appropriate correction for the tectonic processes. On the other hand, the lack of complete time series creates a large time window between the Carboniferous and the Devonian (Torsvik et al., 2012) (60 Ma) which, in the case of using the method of splines, leads to "stretching" of the time span for the Carboniferous era. This definitely suggests the clear conclusion that the pioneering works of Khramov and his colleagues on studying the Carboniferous rocks, primarily, in the Russian Platform, should be continued.

CONCLUSIONS

(1) The new paleomagnetic poles are obtained from the Gzhelian, Moscovian, and Visean rocks of the Carboniferous of the Russian Platform;

(2) The J_n components with inclination-shallowing are revealed in the red rocks of the Gzhelian and Moscovian age, which can be due to the presence of large hematite particles or particle aggregates associated with the interaction of the magnetic and clay grains.

(3) Based on the constructed APWP curve for the EEP, the kinematic parameters of the platform in the Carboniferous are estimated. The total rotation angle is 36° ± 6° with the average velocity of 0.6 degrees per Ma (the rotation pole is φ = 33.3° N, λ = 42.5° E and the latitudinal displacement is 13° ± 7° with an average velocity of 0.2 degrees per Ma (2.4 cm per annum).

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